Citizen Science in Ecology – Study Guide (Group 10)

At a glance

Guiding question: *From new dawn to new standard: how did citizen science develop in ecology?*

Short answer: Citizen science went from an unrecognised side activity (Christmas Bird Count, 1900) to a technology-driven "new dawn" (Silvertown 2009). Ethical and participation frameworks followed (Haklay 2013, ECSA Ten Principles 2015). By 2022 it was a mainstream method with a full project life cycle and quality-control toolkit (Fraisl et al. 2022).

Five things to remember:

1. **Definition.** Citizen science = members of the public actively taking part in scientific research that produces new knowledge. Silvertown: "a volunteer who collects and/or processes data as part of a scientific enquiry."
2. **The old problem never fully went away.** In every era there are doubts about data quality and volunteers' contributions go unrecognised.
3. **Technology changed who can take part.** The internet, smartphones, GPS, cheap sensors and AI apps opened participation to everyone, not just a privileged few.
4. **Frameworks made it credible.** Haklay's 4 levels describe *how deeply* people take part; the ECSA Ten Principles describe *what good practice* looks like.
5. **Quality is now designed in.** Training, community consensus, expert review, automatic filters and statistical correction (MammalWeb, Spipoll, iNaturalist) answer the CBC's old "can we trust this?" question.

Core papers: Silvertown (2009) *A new dawn for citizen science* · Fraisl et al. (2022) *Citizen science in environmental and ecological sciences*. Supporting text: Haklay (2013) typology of participation.

Key concepts

Term	Meaning	Where it comes from
Citizen science	Public participation in scientific research and knowledge production	Silvertown 2009; Fraisl 2022
Contributory citizen science	Designed by professional scientists; volunteers mainly collect data. The focus of Fraisl 2022	Fraisl 2022
Community-led citizen science (CCS)	The public is involved beyond data collection: question, design, interpretation, decisions; scientists advise	Fraisl 2022, Box 1
Crowdsourcing	A large, dispersed group contributes to a task too big for a small group; Haklay's Level 1	Haklay 2013
Extreme citizen science	Haklay's Level 4: citizens and scientists collaborate at every stage; scientists act as facilitators	Haklay 2013; UCL ExCiteS
Opportunistic data	Records made by chance (e.g. on a walk) with no sampling design, so they are uneven and biased	Fraisl 2022
Structured / semi-structured / unstructured	How strictly a protocol fixes where and when sampling happens, and how trained participants are	Fraisl 2022
QA vs QC	Quality *assurance* = before and during collection (training, protocols). Quality *control* = after (flagging outliers, checking photos)	Fraisl 2022
Triangulation	Using several observers, methods and data sources to cancel out single-source bias	Fraisl 2022

Term	Meaning	Where it comes from
Metadata	Data about the data: when, where, how, why it was collected and how quality was checked	Fraisl 2022
FAIR principles	Data should be Findable, Accessible, Interoperable, Reusable	Fraisl 2022
CARE principles	Indigenous data governance: Collective benefit, Authority to control, Responsibility, Ethics	Fraisl 2022
Data minimisation	Collect only the data you truly need (an EU GDPR rule)	Fraisl 2022
Grey literature	Reports, newsletters and similar material not published in peer-reviewed journals	Silvertown 2009
BioBlitz	A short, intense public survey recording as many species as possible in one place (1–2 days)	Silvertown 2009; Fraisl 2022
Ten Principles	ECSA's 2015 good-practice standard for citizen science	ECSA

Timeline: from dawn to standard

The two core papers bracket the rapid modern growth of citizen science

Key milestones, not to scale · highlighted = the two core papers



timeline · key milestones 1600s–2022

Amateurs did most science until the late 1800s. The CBC (1900) began modern mass participation. Silvertown (2009) and Fraisl et al. (2022) frame the period when technology and frameworks turned it into a standard method.

Core paper 1: Silvertown (2009), "A new dawn for citizen science"

Trends in Ecology & Evolution 24(9): 467–471. A short forum article (5 pages). **Main claim:** citizen science is old, but its *modern* form is new because anyone can now take part, and it is exploding in ecology.

1. Origins

- Two centuries ago almost all scientists earned a living in another job. Franklin was a printer and politician; Darwin sailed on the Beagle as an unpaid companion.
- Science became a paid profession only in the **late 19th century**. Amateurs kept going where observation skill matters more than equipment: archaeology, astronomy, natural history.
- **What makes modern citizen science different:** it is "potentially available to all, not just a privileged few."
- Earliest modern example: the **Christmas Bird Count** (since 1900). In the UK the **British Trust for Ornithology** (1932) harnessed amateur birders. The UK National Biodiversity Network then held over 31 million records of over 27,000 species, mostly from amateurs.

2. Three drivers of the "explosion"

1. **Technical tools.** The internet above all, plus mobile computing (CyberTracker, a PDA system first built for non-literate Khoi-San trackers). Also customised electronic field guides showing only local species.
2. **The public as a free resource.** Labour, skills, computing power and even money: Earthwatch volunteers *pay* to help with fieldwork.
3. **Funders demand outreach.** NSF (USA) and NERC (UK) require public engagement; involving people in research is the best outreach.

3. State of the field in 2009

- At the 2008 ESA meeting there were 60+ citizen-science presentations. Yet the ISI Web of Knowledge had only **56 articles** explicitly on "citizen science" (80% from the last 5 years).
- Why under-represented? (a) the term is new, even though hundreds of papers already used CBC data; (b) survey projects don't fit the hypothesis-testing model, so they end up in grey literature.
- His project sample (Table 1) has 3 categories: **hypothesis-driven** (rare: Evolution MegaLab, peppered moth, PigeonWatch), **volunteer mapping and monitoring** (hundreds; e.g. BTO, CBC, OPAL, eBird, Protea Atlas), and **tools and resources** (CyberTracker, iSpot, Citizen Science Toolkit).

4. Three case studies (Box 1)

- **Christmas Bird Count:** about 2,000 circles a year, data free to download, around 350 papers. CBC + Breeding Bird Survey showed that **20 common North American species fell by an average of 68% over 40 years**: "how things look before extinction happens."
- **Evolution MegaLab (2009):** Europe-wide survey of banded snail (*Cepaea*) shell colour. It tests two hypotheses: fewer song thrushes means weaker camouflage selection, and climate warming means fewer dark morphs in the north. It includes an online quiz that trains and tests users (scores used to weight data) and instant feedback against historical samples within 5 km.
- **OPAL (OPen Air Laboratories):** a £12 million Big Lottery programme with 16 institutions. It works with local people, especially disadvantaged groups, on air, soil, water and biodiversity.

5. Principles of good practice he proposes

- Data collected by the public **must be validated** (the only one widely practised then).
- Methods must be **well designed and standardised**.
- Assumptions must be made **explicit**.
- Have a **hypothesis** in mind, even just "how is X changing?"
- Volunteers must get **feedback** as a reward.

Rigour varied hugely, from the BTO's stratified random sampling to schemes accepting records from anywhere.

6. Conclusion of the paper

More scientists should use citizen science for large-scale *research*, not only for education or monitoring. "Science *for* the people" (1970s slogan) should become "**Science *by* the people**," a distinctly 21st-century phenomenon. "Citizen science should not be second-rate science."

Core paper 2: Fraisl et al. (2022), "Citizen science in environmental and ecological sciences"

Nature Reviews Methods Primers 2: 64. A long methods "primer" by 12 authors, including **Haklay**. **Main claim:** contributory citizen science is now a recognised, well-developed *method*, and the paper gives the full how-to, from design to data publishing.

1. Why it has grown

- **Technology:** mobile phones, low-cost sensors, the internet for sharing, cloud storage.
- **Rising literacy and education** worldwide.
- **Open science:** citizen science is a pillar of it. UNESCO's 2019 Recommendation on Open Science was adopted by 193 states.
- Benefits go beyond science: societal (awareness), policy (better laws), political (civic participation), economic, and personal (enjoyment, skills).

2. The six-stage project life cycle (Fig. 1)

The stages are iterative, not strictly one after another.

1. **Identify the need or problem.** Can be done by scientists, participants or both.
2. **Decide if citizen science is the right approach.** It suits wildlife observation, land-cover monitoring, satellite image classification and water or air quality. It is *unsuitable* for expensive or highly technical equipment, or very heavy time demands.
3. **Design the project.** Aims, sampling design, training, communication plan, participant tasks, safety.
4. **Build the community.** Know their motivations, use "citizen science enablers" (facilitators), decide how to credit people, be inclusive.
5. **Manage the data:** plan → collect → assure → analyse → describe and preserve → integrate.
6. **Evaluate.** Front-end, formative (during), summative (end). Then take action (present to decision-makers) and give feedback to participants.

3. Data quality toolkit

- **Training and testing** participants (quizzes, test runs, guides, videos), sometimes used to weight data.
- **Community-based review.** E.g. iNaturalist's "research grade" by community consensus, curators, and participant-experts.
- **Automatic filters** flag records outside expected patterns.
- **Statistical correction.** Correct for observer skill differences and combine opportunistic with structured data (model-based integration).
- **Triangulation** across observers, methods and data sources.
- Expert checking works for small projects but **does not scale** to thousands of participants.

- Growing evidence shows citizen data can match professional data.

4. Example projects in the paper

- **MammalWeb** (UK, camera traps), **Spipoll** (France, pollinators) and the **Barcelona marine guide (URBAMAR)**: biodiversity.
- **FotoQuest Go:** land use and cover across Europe, checked against the EU's LUCAS survey. It paid €1–3 per submission and was GDPR-compliant.
- **Western Redcedar Dieback Map:** climate change and tree health, run on iNaturalist.
- **Community-led monitoring** in Tanzania (Box 1) and **Científicos de la Basura** / Plastic Pirates: school marine-litter science (Box 2).

5. Limitations (Table 4)

- Running a project needs many **non-research skills**: communication, community building, participant management.
- **Engagement** is hard to keep up; unpopular topics attract fewer people.
- **Inclusion:** in unequal societies, only some groups take part.
- **Biases:** dense populations and roads get more samples; days and weekends dominate; a few people supply most of the data; demographics shape coverage.
- **Low-cost sensors** have calibration problems.
- **Different data-protection laws** in different countries (GDPR data minimisation).
- **Safety and political risk.** More than 2,200 people were killed from 2002 to 2020 defending land and the environment.
- **Ethics of open data.** Exact locations of endangered species could help poachers, so coordinates are coarsened.

6. Key numbers to remember

- Citizen science supplies **at least 50%** of observations in global biodiversity databases such as **GBIF**, yet is often unacknowledged.
- At least **33% of the 231 SDG indicators** can be supported by citizen science data.
- It can help fill gaps in **58% of the 93 environment-related SDG indicators**.
- Ghana integrated citizen beach-litter data into official SDG reporting.

7. Outlook / conclusion of the paper

Citizen science is entering the mainstream literature. Barriers remain: low awareness, little career recognition, and hard-to-get funding. The way forward is multi-stakeholder alliances, data aligned with standards, open tools, and integration with official data systems. Legal support is growing: the 1998 Aarhus Convention gives Europeans the right to take part in environmental decisions, and 2021 additions explicitly support citizen science. Grassroots projects (Public Lab after Deepwater Horizon, Safecast after Fukushima) show high-impact work can also come from the bottom up.

Conclusion: from Silvertown's "new dawn" to Fraisl's "new standard"

In one sentence: Silvertown (2009) argued that citizen science *could* be serious science and listed what it would need. Fraisl et al. (2022) show that it *has become* a standard method, with the design, quality-control and ethics machinery that Silvertown asked for now written down and in use.

Question	Silvertown 2009 ("new dawn")	Fraisl et al. 2022 ("new standard")
Type of paper	Short opinion/forum piece	Long methods primer, 12 authors
Status of the field	Barely recognised: 56 articles in Web of Knowledge	Mainstream; regional associations on every continent
Main driver	Internet, PDAs, early smartphones	Smartphones, low-cost sensors, cloud, AI/ML, open science policy
Good practice	"Learning by doing"; only a draft toolkit (Cornell)	Six-stage life cycle; many published guidelines (Table 1); ECSA Ten Principles
Data quality	"Must be validated," usually by experts	Layered QA/QC: training, community consensus, filters, statistics, triangulation; expert-only doesn't scale
Volunteer role	Mostly surveys; few hypothesis-driven projects	Mostly contributory, but the most engaged move to co-creation (MammalWeb)

Question	Silvertown 2009 ("new dawn")	Fraisl et al. 2022 ("new standard")
Feedback / credit	Volunteers "must receive feedback"	Feedback, co-authorship, open data with DOIs, privacy-aware credit
Ethics	Not discussed	Central: GDPR, data sovereignty, CARE, poaching risk, participant safety
Inclusion	"Available to all, not just a privileged few"	Digital divide and demographic bias are recognised problems
Impact on policy	Outreach as public accountability	SDG monitoring (33% of indicators), Aarhus Convention, official data

What *stayed the same*

- **Doubts about data quality.** Silvertown says data "must be validated"; Fraisl still notes that scientists and decision-makers doubt volunteer data.
- **Under-recognition.** In 2009 CBC-based papers weren't labelled citizen science; in 2022 volunteers supply 50%+ of GBIF data but are often unacknowledged.
- **Large scale is the core strength.** Both say large-area, long-term monitoring is impossible without volunteers.

What *changed*

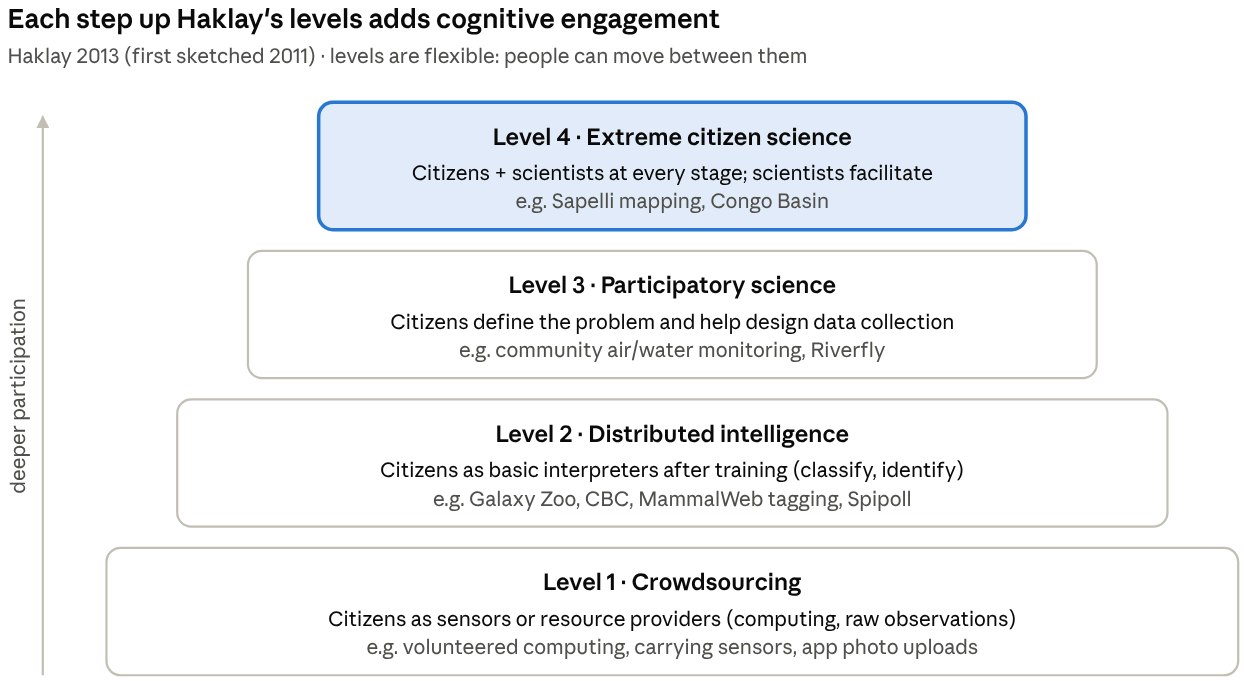
- **Silvertown's wish list became standard practice.** His five principles (validate, standardise, explicit assumptions, hypothesis, feedback) reappear in Fraisl's life cycle and in the ECSA Ten Principles (e.g. #2 genuine science outcome, #5 feedback, #6 limitations and biases).
- **The problem shifted.** It moved from *who can take part* and *are the tools good enough* (2009) to *how do we run it well at scale*: bias, inclusion, privacy, ethics, recognition (2022).
- **Participation deepened.** Silvertown's "science by the people" maps onto Haklay's higher levels; Fraisl's MammalWeb example shows contributors moving towards co-creation.

Take-home message

Citizen science is no longer only about collecting data. It is science done *with* the public: designed with quality control, ethics and feedback built in. Its remaining challenges are about fairness, trust and recognition, not about whether it works.

Haklay's typology: levels of participation

Muki Haklay (geography professor at UCL) published the four-level typology in **2013**. He had sketched it on his blog in 2011. He wanted volunteers recognised as **partners, not just "sensors" or a "crowd"**: many have degrees, lay knowledge and an insider view of local problems. In OpenStreetMap, about half of participants had an undergraduate degree and many had postgraduate degrees.



Haklay's typology · 4 levels of participation

Inspiration: Arnstein's "Ladder of Citizen Participation" (1969). It has 8 rungs, from *manipulation* to *citizen control*, about political power. Arnstein's ladder runs clearly from **bad to good**. Haklay's levels are different: the scientist–volunteer relationship is not a power struggle, **every level has value**, and even minimal engagement has positive outcomes.

Level	Pros	Cons
1 Crowdsourcing	Highly controlled, accurate data; accessible to anyone	Wastes volunteers' thinking (only a problem if they want more)
2 Distributed intelligence	Uses cognitive skills; a test after training shows data quality	Training needed; still scientist-led
3 Participatory science	Local problems tackled; citizens shape the question	Analysis still depends on experts
4 Extreme citizen science	Full collaboration, bottom-up, can even run without professional scientists	Demanding in time and facilitation

Flexible, not fixed: in one project most people may stay at Level 1, committed ones help others (Level 2), and the most engaged discuss results with scientists (Level 3+). Hybrids are normal.

Impact: the typology became shared terminology for describing projects and inspired more flexible frameworks such as the **DITOs Escalator**. Major scientific bodies adopted it to evaluate proposals, and grant makers began asking applicants to state their intended level. Haklay went on to co-direct **UCL Extreme Citizen Science (ExCiteS)** and co-author the **ECSA Ten Principles** and Fraisl et al. 2022.

Where the presentation's examples sit: CBC and most apps at 1–2 · Spipoll at 2 · MammalWeb at 2, with its co-creation spin-offs moving to 3–4.

ECSA Ten Principles of Citizen Science

The **European Citizen Science Association (ECSA)** wrote them in **2013–2015**. Citizen science had grown fast in the early 2010s with no guidance. The aims were to **set good practice, provide a framework and build credibility**. Haklay was one of the co-authors.

#	Principle	In short	Example from the presentation
1	Active citizen involvement	Citizens take a meaningful role: contributor, collaborator or project leader	MammalWeb volunteers deploy cameras and classify images
2	Genuine science outcome	Answers a research question or informs conservation, management or policy	MammalWeb data led to a local nature reserve
3	Mutual benefits	Both scientists and volunteers gain: publications, learning, enjoyment, social benefits, policy influence	MammalWeb improved students' mental health during COVID-19
4	Participation in multiple stages	Volunteers *may*, if they wish, help with question, method, data, analysis, communication	MammalWeb spin-off projects led by volunteers
5	Participant feedback	Volunteers learn how their data are used and what came of it	Spipoll newsletters, weekly news, annual meetings
6	Limitations and biases acknowledged	A research method like any other, with biases to control; also democratises science	CBC observers favouring the "best spots"
7	Open data and metadata	Data public and results open access where possible, unless privacy or security forbids	Spipoll open-access datasets
8	Proper acknowledgment	Volunteers credited in results and publications	Credit / co-authorship in Fraisl 2022

#	Principle	In short	Example from the presentation
9	Programmatic evaluation	Evaluated for science output, data quality, participant experience and societal/policy impact	Fraisl's Stage 6
10	Legal and ethical considerations	Copyright, IP, data sharing, confidentiality, attribution, environmental impact	MammalWeb removes human images; locations coarsened

Memory hook: 1–5 are about **people and process** (involve, outcome, benefit, stages, feedback); 6–10 are about **rigour and responsibility** (biases, open data, credit, evaluation, law and ethics).

Case studies

Christmas Bird Count (CBC): the "dawn"

- **What:** an annual bird census. Volunteers (mostly North America, also Latin America and the Caribbean) count every bird they see or hear in a fixed **count circle of 15 miles (24 km) in diameter**, within **24 hours**, during the Christmas period.
- **Who runs it:** the **National Audubon Society** (US non-profit). Over **85,000 participants a year**, running for **more than 125 years**: the longest-running citizen science survey in the world.
- **Origin:** before 1900, the Christmas **"Side Hunt"** was a family competition to shoot the most birds and small mammals. On Christmas Day **1900**, ornithologist **Frank M. Chapman** proposed a "Christmas Bird Census" instead. Reports were mailed to the magazine *Bird-Lore* with locality, times, temperature, weather, wind and counts per species.
- **Why it matters today:** population trend detection, climate-change documentation (shifting winter ranges), species-specific conservation, public decision-making. Silvertown: about 350 papers, and evidence of a 68% average decline in 20 common species.
- **Data problems ("Was the data reliable?"):**
  - No direct way to test data quality except agreement with expected patterns or independent sources.

    - Before modern tech, all data were gathered and compared by hand.
    - Contest spirit: rare birds staked out, seed scattered, observers spend time in the best spots. So **rare species appear more abundant** than they are.
- **How it handles this now:** teams follow assigned routes or feeders inside the circle. Volunteer **compilers** recruit and train people, ensure habitat coverage, verify counts and submit them through Audubon's system. Two ways to take part: **feeder counts** and **field counts** (some include night surveys for owls). Analyses are **adjusted for observer effort**.
- **Haklay level:** mostly 1–2 (data collection, some identification skill).

MammalWeb (UK and Europe): the standard in practice

- **Set up in 2013** by **Durham University** and **Durham Wildlife Trust**; major funding from **2015**. It was run by MammalWeb Limited until 2025 and became a registered charity at the end of 2024.
- **Problem it solves:** mammals are elusive, nocturnal and few, so national-scale monitoring without volunteers would be very costly.
- **Workflow (the slide's 8 keywords):**
    1. **Camera-trapping:** volunteers deploy their own or borrowed motion-triggered cameras.
    2. **Centralised upload** with time and location metadata.
    3. **Privacy scrubbing:** images of humans are removed as soon as they are flagged (Principle 10).
    4. **Crowdsourced tagging:** registered users classify the animals.
    5. **Consensus verification:** several classifications per sequence, plus an expert-classified subset, combined into **consensus classifications with confidence levels** (answers the CBC's quality problem).
    6. **Machine-learning automation:** the tagged dataset trains AI for automatic species recognition.
    7. **Co-creation transition:** the most engaged volunteers start their own projects (contributory → co-created).
    8. **Nature reserve designation:** a volunteer-started County Durham survey (2018) led to a **local nature reserve** (Principle 2).
- **Numbers (Fraisl 2022):** 270+ participants in 8 countries, 50+ schools, 20+ organisations, over 340 years of camera time, 620,000+ sequences, about 2 million photos. Helped locate potentially invasive raccoons and coati. Improved student

mental health during COVID-19.

- **NHMP, National Hedgehog Monitoring Programme:** a hedgehog project that uses **camera traps on the MammalWeb platform**. **The two graphs on the slide ("All classifications by species" and "Sequences uploaded and classified, Oct 2025–Oct 2026") come from MammalWeb. Camera traps photograph every animal that passes, so common species dominate the classifications.** Livestock (~390k), grey squirrel (~270k) and pheasant (~220k) are the most-classified named categories; uploads grew from about 2.8 to 5.2 million sequences in a year.
- **Haklay level:** 2, moving towards 3–4 through co-creation.

Spipoll (France, 2010)

- **Spipoll** = *Suivi photographique des insectes pollinisateurs* (photographic survey of flower visitors). Launched in **2010** by the **French National Museum of Natural History (MNHN)** and **OPIE** (Office for Insects and their Environment), as part of Vigie-Nature.
- **Protocol** (no prior insect knowledge needed):
  1. **Site selection:** any flowering plant, from city centre to natural habitat.
  2. **Standardised observation:** photograph every invertebrate landing on its flowers for **20 minutes**.
  3. **Identification** of insects and plant with a dedicated online tool (step-by-step key).
  4. **Upload:** photos, IDs, date, time, location, weather.
  5. **Quality control:** originally experts only; **since 2019 also peer validation** by participants, plus entomologist review. It also uses **statistical weighting**: species often misidentified get lower confidence and are checked by experts more often. It also filters out human images for privacy.
  6. **Community engagement:** comments on others' observations (a social network), annual meetings, weekly news, monthly newsletter (Principle 5).
  7. **Analysis and open-access publication:** insect–flower networks vs. urbanisation and land use.
- **Findings:** effects of urbanisation on pollinator communities, different land-use affinities, **domestic gardens as good pollinator habitat**.
- **Numbers (slide):** 5,627 participants, 97,605 collections.
- **Privacy note (both projects):** exact locations are collected, but published datasets use **lower spatial resolution**.
- **Haklay level:** 2 (collection + identification + peer validation).

Mobile tools: AI in the volunteer's pocket

All four apps lower the skill barrier, which was Silvertown's paper field-guide problem. They mostly keep users at **Haklay Level 1–2**.

App	Made by	Identifies	How it works	Quality / science link
Merlin Bird ID	Cornell Lab of Ornithology	Birds by sound, photo or questions	Sound ID listens live and works offline (2,066+ species in US, Canada, Europe, India, Neotropics); Photo ID; Step-by-Step ID (place, date, size, colour, behaviour)	Life List synced with **eBird**
BirdNET	Cornell Lab (K. Lisa Yang Center for Conservation Bioacoustics) + Chemnitz University of Technology	6,000+ bird species by sound	Audio cut into **3-second clips** → **Mel-spectrograms** → **convolutional neural network (CNN)** predicts species → filtered by location and date	Open source
Pl@ntNet	French institutes (Cirad, INRAE, Inria, IRD) + Tela Botanica	Plants from photos (~20,000 of ~360,000 species per the slides; focus on wild flora)	Image recognition; filter by family or genus; multiple regional floras	Photos shared with scientists; expert community validates

App	Made by	Identifies	How it works	Quality / science link
iNaturalist	California Academy of Sciences + National Geographic Society	Plants, animals, fungi	Instant AI camera suggestions; multiple photos or sound	**Research grade** via community consensus (400k+ scientists and naturalists); GPS automatically recorded; used in Fraisl's Western Redcedar project

Exam angle: these tools show the "new tools & data streams" of the topic title. They solve *access* and *identification*, but bring Fraisl's new problems: digital divide, opportunistic (biased) data, privacy of GPS data.

Challenges across three eras

The concerns moved from **who could take part**, to **whether tools and methods were good enough**, to **how to run citizen science well at scale**. Two problems run through all three eras: **data-quality doubts** and **under-recognised volunteers**.

Era	Main challenges	Source
Pre-technology (before ~2000)	Elite access (a privileged few); only fields not needing costly equipment; confusing, costly paper field guides; hard to coordinate many observers (CBC circles, effort corrections); work unrecognised or left in grey literature	Silvertown 2009
Post-technology (~2009)	Immature software that quickly goes out of date; no standard methods ("learning by doing"); inconsistent rigour; few hypothesis-driven projects; needs partner coalitions and shared infrastructure; low academic recognition (56 articles)	Silvertown 2009

Era	Main challenges	Source
Modern (2022)	Non-research skills needed; participant engagement and retention; inclusion and digital divide; sampling biases (population, roads, weekends, a few heavy contributors, demographics); scaling quality control and trust; low-cost sensor calibration; privacy, GDPR, data sovereignty, CARE; poaching risk from open locations; credit vs. privacy; safety and political risk; little funding or career recognition	Fraisl et al. 2022

Pros and cons in one view

Pros	Cons
Huge spatial and temporal scale no professional team can afford	Data reliability questioned
Fills data gaps (50%+ of GBIF records)	Sampling and observer biases
Public engagement, education, democratised science	Engagement hard to sustain
Feeds policy (SDGs, nature reserves, waste laws)	Privacy, safety and ethical risks
Local and Indigenous knowledge included	Excludes people without tech access

Self-test

Cover the answer column and try each one first.

#	Question	Answer
1	What, according to Silvertown, separates modern citizen science from its historical form?	It is potentially available to **all**, not just a privileged few

#	Question	Answer
2	Name Silvertown's three drivers of growth.	Technical tools (internet, mobile); public as free labour, skills, computing and finance; funders requiring outreach
3	Why was citizen science under-represented in the literature in 2009?	The term was new, and survey projects don't fit the hypothesis-testing model, so they ended up in grey literature
4	Which of Silvertown's good-practice principles was widely applied in 2009?	Validation of volunteer data (usually by experts)
5	What kind of citizen science does Fraisl et al. 2022 focus on, and why?	Contributory: a manageable entry point with many examples
6	List Fraisl's six project stages.	Identify need → check if CS fits → design → build community → manage data → evaluate
7	QA vs QC?	QA before and during collection (training, protocols); QC after (outlier flags, photo checks)
8	Why doesn't expert-only checking work for big projects?	It doesn't scale to thousands of participants; it needs community consensus, filters and statistics
9	Give four sampling biases from Fraisl.	Population density, roads, daytime/weekends, a few heavy contributors (also demographics)
10	Share of GBIF observations from citizen science? Share of SDG indicators supportable?	At least 50%; at least 33% of 231 indicators
11	Why was the CBC created?	To replace the Christmas "Side Hunt": Frank Chapman, 1900
12	Why might rare species look too common in CBC data?	Contest spirit: observers stake out rare birds and favour the best spots
13	Haklay's four levels, low → high?	Crowdsourcing → distributed intelligence → participatory science → extreme citizen science

#	Question	Answer
14	How does Haklay's typology differ from Arnstein's ladder?	Arnstein = bad → good political power; Haklay = every level has value, levels are flexible and people can move between them
15	Who wrote the Ten Principles, when, and why?	ECSA, 2013–2015: good practice, framework, credibility
16	How does MammalWeb ensure data quality?	Multiple classifications + expert subset → consensus with confidence levels; ML trained on it
17	What is Spipoll's protocol?	Photograph all flower visitors on one plant for 20 minutes; identify with online key; upload; peer and expert validation
18	Which Ten Principle does removing human images from MammalWeb serve?	#10 Legal and ethical considerations
19	One thing that stayed the same from 2009 to 2022?	Doubts about data quality and under-recognition of volunteers
20	One thing that changed?	From ad-hoc "learning by doing" to a standard life cycle with built-in QA/QC, ethics and open data

Extra materials and links

Core and supporting papers

- Silvertown, J. (2009). A new dawn for citizen science. *Trends in Ecology & Evolution* 24(9): 467–471.
- Fraisl, D. et al. (2022). Citizen science in environmental and ecological sciences. *Nature Reviews Methods Primers* 2: 64.
- Haklay, M. (2013). Citizen Science and Volunteered Geographic Information: overview and typology of participation (DOI)
- Arnstein, S. (1969). A Ladder of Citizen Participation
- ECSA (2015). Ten Principles of Citizen Science

Haklay and frameworks

- Haklay: Citizen science as participatory science (blog, 2011)
- Haklay: Classification of citizen science activities (2011)
- Haklay: Participatory citizen science (2016)
- UCL: Types of citizen science
- UCL: History of citizen science
- UCL Extreme Citizen Science (ExCiteS)
- DITOs: Doing It Together Science
- Citizen science impact pathways (CSTP)
- Citizen Science Terminology Matters

Christmas Bird Count

- Audubon: About the CBC
- Audubon: Join the CBC
- Audubon: CBC turns 125
- Hawaii Audubon: CBC (circle size)
- Chapman's original 1900 *Bird-Lore* proposal (HathiTrust)
- Studies in Avian Biology: CBC data analysis
- History of the Christmas Bird Count

Projects

- MammalWeb
- MammalWeb and the 10 principles of citizen science
- Spipoll
- Spipoll on Wikipedia (French)
- Video used in the presentation (YouTube)
- Northeastern: citizen science improvements

Apps to try

- Merlin Bird ID · BirdNET · Pl@ntNet · iNaturalist · eBird

Platforms named in Fraisl 2022 (Table 2)

- CitSci.org · Zooniverse · iRecord · EpiCollect · Sensor.Community · GBIF